Software Engineering Documentation - StartupScan

Complete and detailed version

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1. Scope and objectives

StartupScan is an AI-powered startup evaluation platform, aimed at pitch validation, executive communication, and decision support. The scope covers multimodal processing, scoring, report generation, AI video generation, visual pitch decks, and model management.

Operational objectives:

• Reduce the time needed to analyze a startup opportunity.

• Standardize technical and investment feedback.

• Generate artifacts ready for stakeholder meetings.

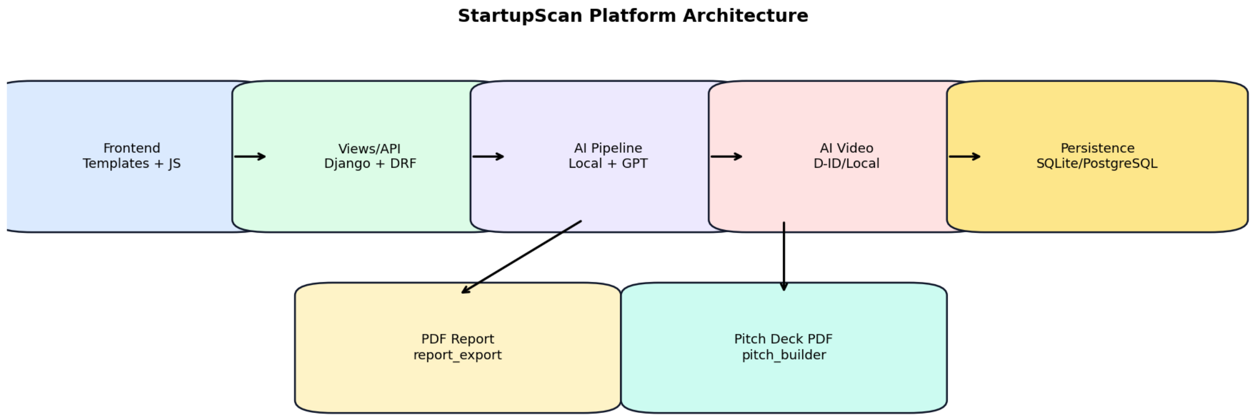
• Maintain traceability of analyses and model iterations.

2. Implemented requirements

Functional requirement	Status	Notes
Multimodal submission (text/doc/audio/video/youtube)	Implemented	Main flow in the pitch form
Local/GPT evaluation with fallback	Implemented	Automatic fallback for resilience
Score 0-10 + categories + recommendations	Implemented	With an interpretable block for investors
Technical PDF report of the analysis	Implemented	Available on the result page
AI video (auto/did_only/local_only)	Implemented	Asynchronous with a progress endpoint
Slide-based pitch PDF	Implemented	Visual deck with a professional layout
Automatic + manual premium design in the pitch PDF	Implemented	Configurable templates
Model management (training/retraining/activation)	Implemented	With real-time progress
Operational and investor dashboard	Implemented	With charts and filters

3. Technical architecture

The architecture consists of a server-side rendering frontend (Django templates), a Django/DRF backend, specialized AI services, and a relational persistence layer.



Stack:

• Backend: Django + DRF

• AI: scikit-learn, OpenAI SDK

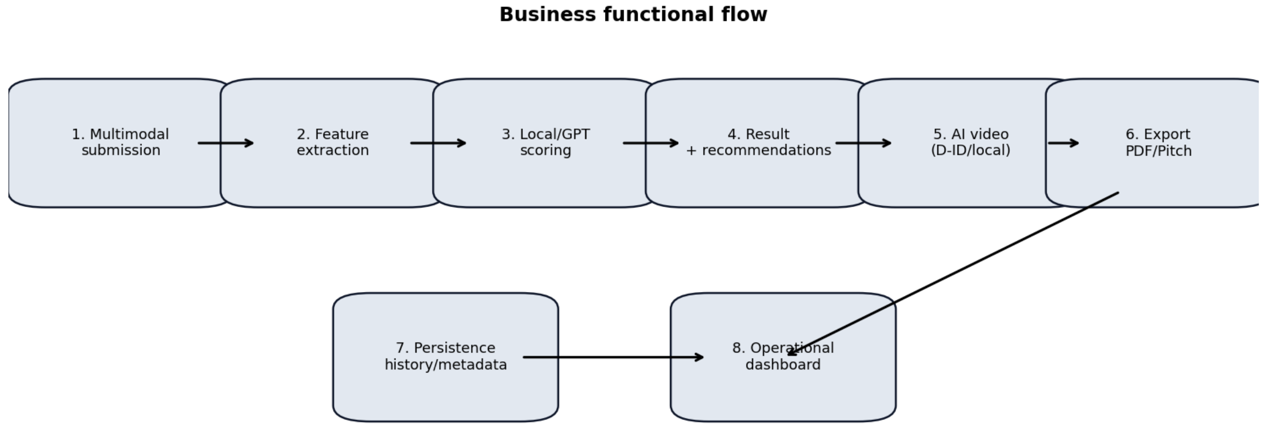
• Video/audio: moviepy, edge-tts, gTTS, D-ID API

• PDF/docs: reportlab, pypdf, python-docx

• Frontend: Bootstrap + JS + Chart.js

• Database: SQLite (dev), PostgreSQL (compatibility)

4. Business flows



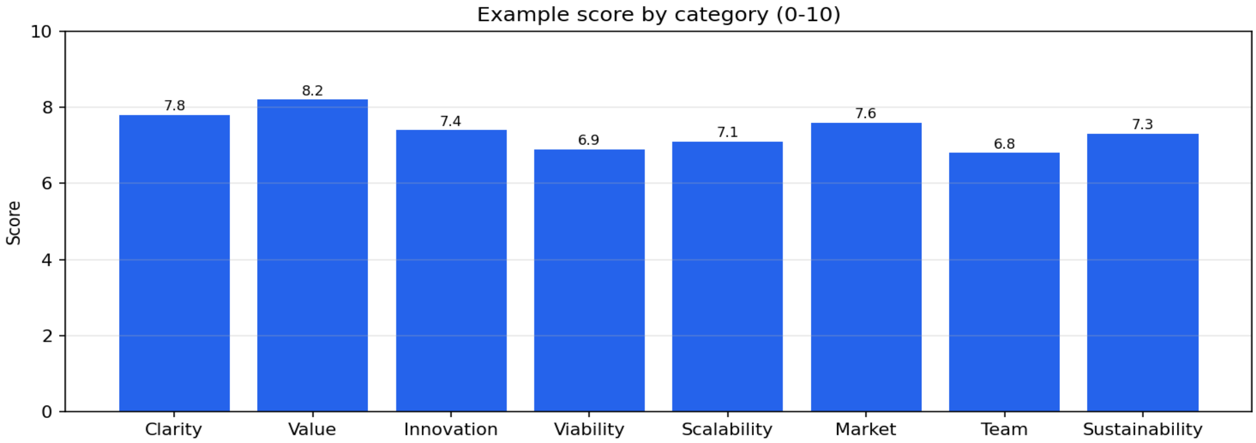
The flow starts with multimodal submission, goes through feature extraction and AI inference, and ends with decision artifacts: report, video, and pitch deck.

5. Data model and persistence

Entity	Responsibility	Critical fields
PitchAnalysis	Main evaluation record	startup_name, score, report, metadata, multimodal files
IdeaPitchSubmission	Idea-to-full-pitch flow	startup_name, problem, solution, generated_pitch, status

The JSON metadata layer is used to attach dynamic information about jobs, generation modes, narrative uniqueness keys, and processing state.

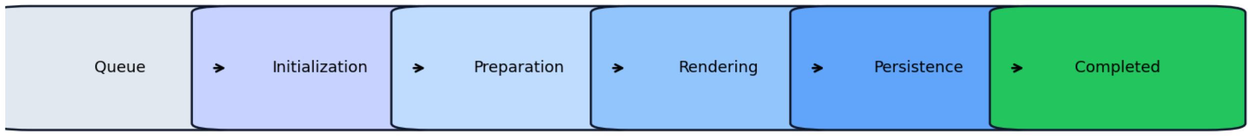
6. Evaluation and interpretability pipeline



- Extraction and consolidation of multimodal context.
- Score prediction and interpretable report generation.
- Standardized categories to facilitate comparison between startups.
- Recommendations oriented toward action and investment readiness.

7. AI video pipeline

Asynchronous job phases (video/training)



Video generation runs as an asynchronous job with phase-based progress. It supports the auto, did_only, and local_only modes, with detailed error handling per scenario and mandatory completion at the end.

Mode	Description	Fallback
auto	Attempts realistic D-ID and falls back to local when needed	Yes
did_only	Forces exclusive use of the D-ID API	No
local_only	Local rendering with no external dependency	Not applicable

8. Pitch deck PDF pipeline

The pitch deck is exported in slide-based format (one page per slide), with two design strategies: automatic by context and manual premium by template.

Design mode	Behavior	Templates
automatic by context	Selects the visual identity based on the startup's context	orbit/grid/wave/diagonal/aurora/ribbon
manual premium	The user explicitly chooses the template	orbit/grid/wave/diagonal/aurora/ribbon

9. Main APIs and routes

Route	Method	Description
/analyze/form/	GET/POST	Multimodal submission and analysis
/results/<id>/	GET	Full analysis view
/results/<id>/pdf/	GET	Technical PDF report
/results/<id>/pitch/pdf/	GET	Visual pitch deck PDF
/results/<id>/video/generate/	POST	Starts an AI video job
/results/<id>/video/progress/<job_id>/	GET	Video job status and progress
/models/	GET/POST	Model management and training
/investors/	GET	Investor-oriented dashboard

10. Operational guide

1. Start the application and check basic health via the dashboard.
2. Run a multimodal submission with financial data.
3. Check score, categories, and recommendations in the result.
4. Generate a video in auto mode and track progress.
5. Generate a pitch PDF in automatic mode and manual premium mode.
6. Download the report and check the format for stakeholders.
7. Monitor logs and analysis metadata for traceability.

11. Security, reliability, and fallbacks

- Format validation and upload exception handling.
- Normalized error messages for the frontend.
- Local fallback when GPT/D-ID is unavailable (where applicable).
- Error separation by scenario for quick diagnosis.
- User-based access control on sensitive routes.

12. Recommended testing and validation

- python3 manage.py check
- python3 manage.py test
- python3 docs/generate_engineering_pdf.py
- python3 docs/generate_engineering_docx.py

Minimum functional scenario: multimodal submission, result with score, AI video with progress, pitch deck PDF in both design modes, and technical report download.

13. Documentation delivery and publishing

The operational routine includes publishing the updated documentation (PDF and DOCX) to the project's Discord webhook, ensuring immediate distribution to the team and stakeholders.